

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech

Branch :Petrochemical Engineering

Semester : VI

Subject Code & Name: BTCHC601_Y21 Mass transfer Operation -II

Max Marks: 60

Date:18/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 For a distillation column operating at minimum reflux, the		1
a. Concentration of liquid and vapor leaving a plate will be same		
b. Reflux ratio will be maximum		
c. Number of plates required will be maximum		
d. None of the above		
2 The equilibrium liquid composition compared to the vapor composition in case of azeotropic mixture is		1
a. More b. Less c. Same d. Either more or less; depends on the system		
3 Dry bulb temperature of the gas is _____ the wet bulb temperature.		1
a. Less than b. More than c. Equal to d. None		
4 Air initially at 101.3 kPa and 40°C and with a relative humidity of 50%, is cooled at constant pressure to 30°C. The cooled air has a		1
a. Higher dew point b. Higher absolute (specific) humidity		
c. Higher relative humidity d. Higher wet bulb temperature		
5 Drift in a cooling tower is		1
a. The water entrained by the circulating air		
b. Dependent on the water lost by evaporation		
c. Desirable		
d. All of the above		
6 The term "cooling range" in a cooling tower refers to the difference in the temperature of		1
a. cold water leaving the tower and the wet bulb temperature of the Surrounding air.		
b. hot water entering the tower and the wet bulb temperature of the surrounding air.		
c. hot water entering the tower and the wet bulb temperature of the surrounding air.		
d. None of the above		

- 7 A slurry is to be dried to produce flaky solid. Which dryer would you recommend? 1
 a. Spray dryer b. Drum dryer c. Rotary dryer d. Dielectric dryer
- 8 Crystallization is based on the _____ 1
 a. Difference in melting point b. Difference in boiling point
 c. Difference in pressure d. Difference in solubility
- 9 The nature of the crystallization process is governed by----- 1
 a. Thermodynamics b. Kinetic factors
 c. Thermodynamics and Kinetic factors d. None of the above
- 10 What is the driving force in Dialysis? 1
 a. Pressure difference b. Pervaporation
 c. Difference in fugacity d. Concentration difference
- 11 The selection of membrane does not depend on which property? 1
 a. Pore size distribution b. Water permeability
 c. Perporometry d. Film thickness formed
- 12 Which of the following is not separated through the distillation process? 1
 a. Acetone and water b. Milk and water
 c. Impurities in Sea water d. Aniline and chloroform.
- Q. 2 Solve the following.** 12
- A) Explain relative volatility. Show that VLE data $y = \alpha x / (1 + (\alpha - 1) x)$ for ideal gas phase binary system, where α is relative volatility. Understanding/ applying 6
- B) Explain Simple distillation and Flash distillation with diagram in detail. understanding 6
- Q.3 Solve the following.** 12
- A) Explain the terms using material balance diagram of distillation for a) Total condenser b) Partial condenser and c) Reboiler Understanding/ applying 6
- B) A continuous fractionating column is to be designed for separating 10,000 kg per hour of a liquid mixture containing 40 mole percent methanol and 60 mole percent water into an overhead product containing 97 mole percent methanol and a bottom product having 98 mole percent water. A mole reflux ratio of 3 is used. Calculate number of ideal plates if the feed is at its bubble point. Analyze 6
- | | | | | | | | | | |
|---|-------|-------|-------|-------|------|-------|-------|-------|------|
| x | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 |
| y | 0.417 | 0.579 | 0.669 | 0.729 | 0.78 | 0.825 | 0.871 | 0.915 | 0.96 |
- Where x = mole fraction of methanol in liquid
 And y = mole fraction of methanol in vapor
- Q. 4 Solve Any Two of the following.** 12
- A) Moist air exists at 40°C dry-bulb temperature, 20°C thermodynamic wet-bulb temperature, and 101.325 kPa pressure. Determine the humidity ratio, enthalpy, dew-point temperature, relative humidity, and specific volume. (psychrometric chart is given at end.) Analyze 6

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|----|--|---------------|---|
| B) | Define dry bulb temperature and wet bulb Temperature. Describe wet bulb temperature in detail with neat diagram. | Remembrance | 6 |
| C) | How the cooling towers are classified? Describe any one type of cooling tower with neat diagram. | Understanding | 6 |

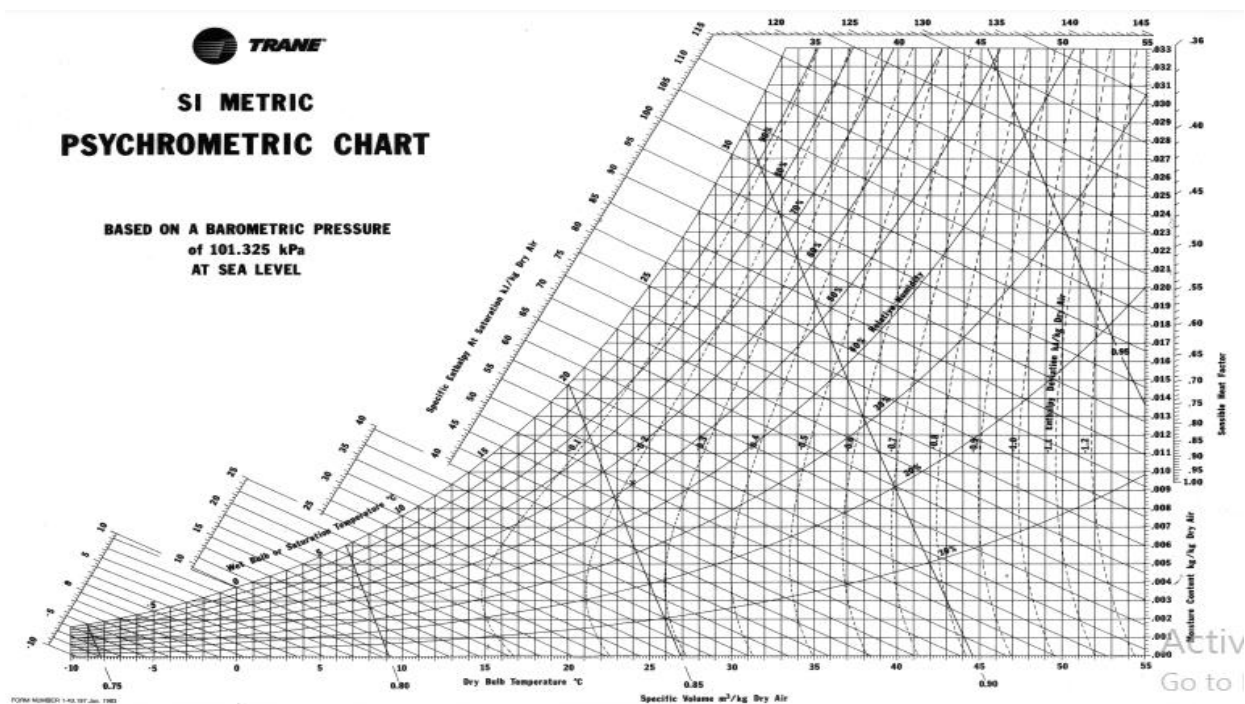
Q.5 Solve Any Two of the following. 12

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|----|---|-----------------------------|---|
| A) | Find an expression for obtaining the total drying time for the drying of a wet solid material under constant drying conditions of to a final moisture content well below the critical moisture content. | Understanding | 6 |
| B) | Describe any one type of drier with neat diagram | Remembrance | 6 |
| C) | What are the parameters controlling the crystal size distribution in a crystallizer? Explain them briefly. | Remembrance / understanding | 6 |

Q. 6 Solve Any Two of the following. 12

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|----|--|---------------|---|
| A) | With neat flow sheet explain RO water treatment plant | Understanding | 6 |
| B) | Write short note on a) Pervaporation b) Membrane Materials | Understanding | 6 |
| C) | Classify the different membranes separation processes based on different driving force and separation size range | Analyzing | 6 |

*** End ***



Psychrometric Chart